

In Silico Mutagenesis and Docking

Goal: Generate a mutant protein model, clean the structure, and calculate binding affinity with a ligand using AutoDock Vina.

PHASE 1: Structural Preparation and Mutagenesis (PyMOL)

This phase takes a raw PDB file and converts it into a docking-ready, clean mutant model.

1.1. Load Structure

Open PyMOL and fetch the structure using its PDB code:

```
fetch 8VYK
```

1.2. Deep Cleaning

Remove water, ions, sugars, and detergents that will interfere with docking:

```
remove solvent
remove inorganic
remove organic
```

Note: If specific glycans or detergents remain, remove by residue name, e.g.: remove resn NAG

1.3. In Silico Mutagenesis

Visualise the target residue:

```
select target, resi 347
zoom target
show sticks, target
color red, target
```

Then apply the mutation:

- Go to Menu: Wizard → Mutagenesis → Protein
- Click on the residue to be changed in the viewer
- In the right panel, click 'No Mutation' and select the new amino acid (e.g., LYS for N→K)
- Click Apply, then Done

1.4. Extract Grid Box Coordinates

To define the center of the docking search box, retrieve the CA coordinates of the mutated residue:

```
iterate_state 1, (resi 347 and name CA), print(x,y,z)
```

Note down the three numbers (X, Y, Z) that appear in the log. These will be used as center_x, center_y, center_z in the Vina config file.

1.5. Save Clean File

```
save receptor_clean.pdb
```

PHASE 2: File Conversion for Docking

2.1. Ligand Preparation — Neutral Form

Use OpenBabel to convert a SMILES code or SDF file to PDBQT at pH 7.4:

```
obabel -:"[SMILES_CODE]" -O ligand.pdbqt --gen3d -p 7.4
```

Or from an existing SDF file:

```
obabel ligand.sdf -O ligand.pdbqt --gen3d -p 7.4
```

2.2. Ligand Preparation — Protonated Form (+1)

To generate the physiologically correct protonated form of mexiletine:

```
obabel ligand.sdf -omol2 -O ligand_protonated.mol2 -p 7.4
```

Verify protonation state — the nitrogen should appear as N.4 (ammonium) in the mol2 file:

```
grep 'N' ligand_protonated.mol2 | head -5
```

Note: N.4 atom type and formal charge +1 confirm correct protonation. If N.3 appears instead, the molecule is neutral.

Convert to PDBQT for Vina:

```
obabel ligand_protonated.mol2 -O ligand_protonated.pdbqt
```

2.3. Receptor Preparation (AutoDockTools)

Launch AutoDockTools:

```
conda activate adt_env && adt
```

Then follow these steps in the ADT interface:

- Load: File → Read Molecule → Select receptor_clean.pdb
- Hydrogens: Edit → Hydrogens → Add (Select 'Polar only') → OK
- Charges: Edit → Charges → Compute Gasteiger
- Save: Grid → Macromolecule → Choose → Select receptor_clean → Save as receptor.pdbqt

PHASE 3: Execution (AutoDock Vina)

3.1. Create Configuration File (Site-Specific Docking)

Create a file named config.txt in the project folder with the following content. Replace the coordinate values with those obtained in step 1.4:

```
receptor = receptor.pdbqt
ligand = ligand_protonated.pdbqt
center_x = [X_COORDINATE]
center_y = [Y_COORDINATE]
center_z = [Z_COORDINATE]
size_x = 26
size_y = 26
size_z = 26
exhaustiveness = 128
num_modes = 9
energy_range = 3
out = results.pdbqt
log = log.txt
```

Note: exhaustiveness = 128 provides high precision but takes longer. size 26×26×26 Å is suitable for a targeted site. For blind docking, increase box size to cover the full protein (see section 3.2).

3.2. Create Configuration File (Blind Docking)

For blind docking, the box must cover the entire protein. First calculate the protein center and dimensions:

```
python3 -c " import MDAnalysis as mda u = mda.Universe('receptor_clean.pdb')
pos = u.atoms.positions print('Center:', pos.mean(axis=0).round(1))
print('Size X:', (pos[:,0].max()-pos[:,0].min()).round(1)+10) print('Size
```

```
Y:', (pos[:,1].max()-pos[:,1].min()).round(1)+10) print('Size Z:',
(pos[:,2].max()-pos[:,2].min()).round(1)+10) "
```

Then create blind_config.txt:

```
receptor = receptor.pdbqt
ligand = ligand_protonated.pdbqt
center_x = [PROTEIN_CENTER_X]
center_y = [PROTEIN_CENTER_Y]
center_z = [PROTEIN_CENTER_Z]
size_x = [PROTEIN_SIZE_X + 10]
size_y = [PROTEIN_SIZE_Y + 10]
size_z = [PROTEIN_SIZE_Z + 10]
exhaustiveness = 128
num_modes = 20
energy_range = 5
out = blind_docking_out.pdbqt
log = blind_docking_log.txt
```

3.3. Run Vina

```
vina --config config.txt
```

PHASE 4: Result Analysis

4.1. Read the Score

Look at the terminal output or the log.txt file. The affinity (kcal/mol) in Mode 1 is the best predicted binding energy. More negative = stronger binding.

4.2. Save Best Pose as mol2

Extract pose 1 from the docking output and save as mol2 (preserving atom types):

```
obabel results.pdbqt -omol2 -O pose1.mol2 -f 1 -l 1
```

Note: For protonated ligand: verify N.4 atom type is preserved in the mol2 file after extraction.

4.3. Build Complex PDB

In PyMOL, combine the clean receptor with the best docking pose:

```
load receptor_clean.pdb, receptor
load pose1.mol2, mex
create complex, receptor or mex
save complex_final.pdb, complex
```

Then fix chain ID and atom type issues for CHARMM-GUI compatibility:

```
sed -i 's/UNL 1/UNL A 1/g' complex_final.pdb
sed -i 's/N1+/N /g' complex_final.pdb
```

4.4. Visualization

Inspect how the ligand fits into the binding site:

```
load receptor_clean.pdb
load results.pdbqt
zoom (resi 347 or resi 348 or resi 349 or resi 353 or resi 355), 8
```